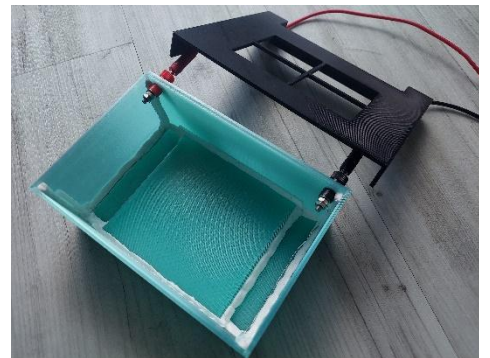
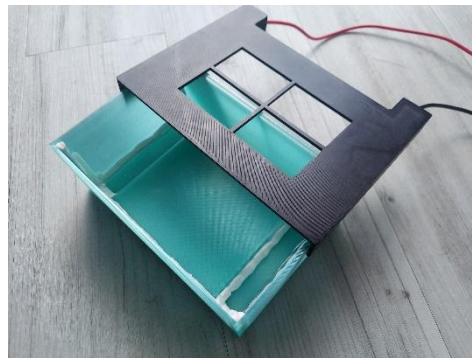
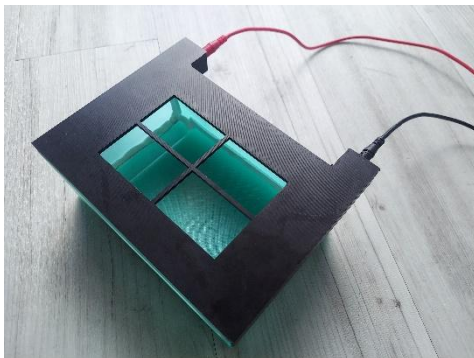
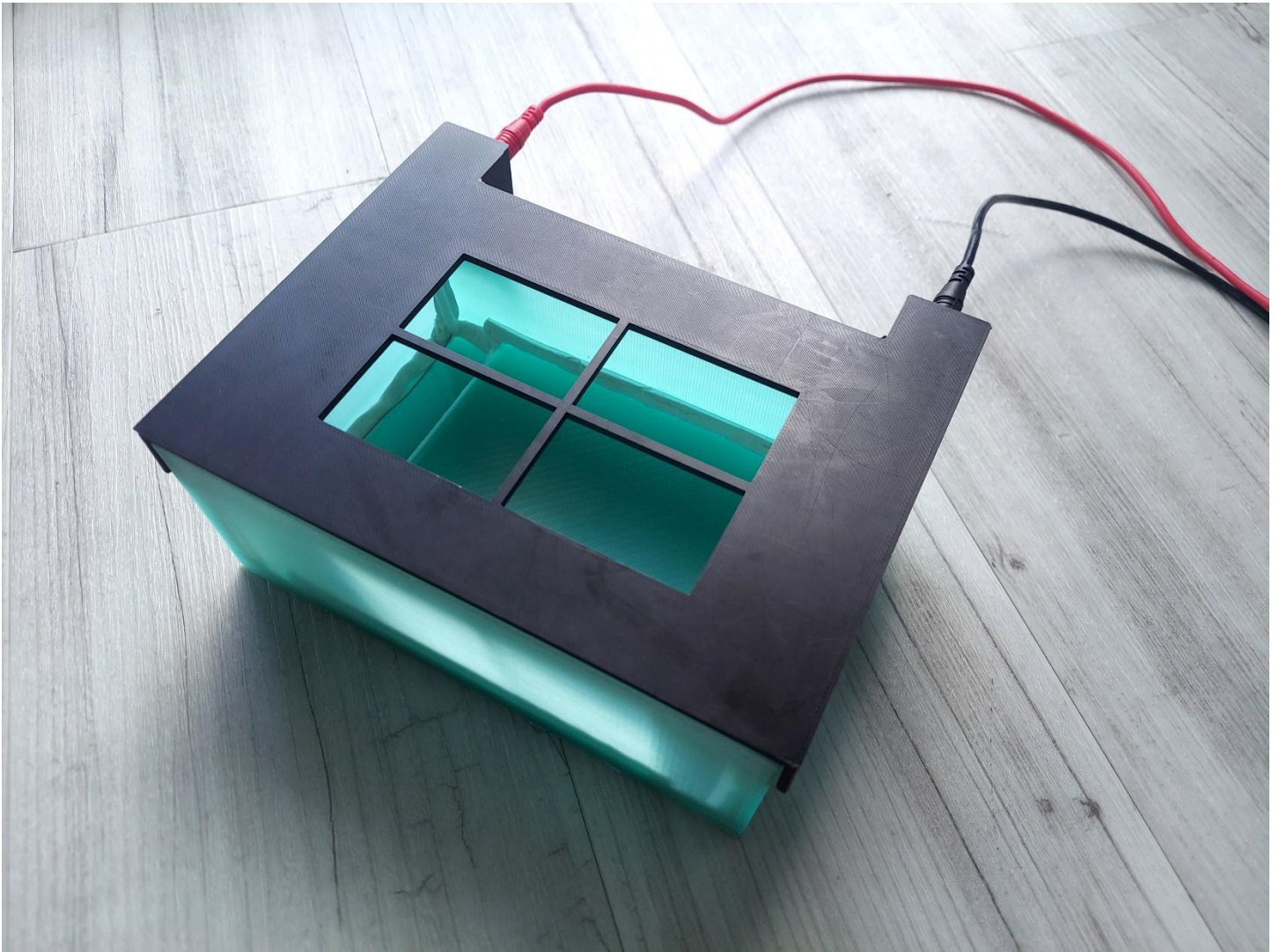


EZ-Phoresis Assembly Instructions

Version 1

Date: September 10, 2026



Safety Warning and Disclaimer

Electrical Hazard

WARNING: EZ-Phoresis is an electrically powered gel electrophoresis device. Electrophoresis power supplies can generate voltages capable of causing electrical shock, serious injury, or death.

Users must understand the electrical and laboratory hazards associated with electrophoresis before constructing or operating this device.

Required Safety Practices

- Always operate the device with the protective lid fully installed.
- Never touch the electrophoresis buffer, electrodes, conductive components, or internal wiring while the system is energized.
- Disconnect the power supply before removing the lid or making any adjustment to the device.
- Inspect the device, electrodes, wiring, connectors, and lid before each use.
- Do not operate a damaged, leaking, improperly assembled, or electrically compromised device.
- Use only a suitable electrophoresis power supply and appropriately rated electrical components.
- Follow applicable institutional laboratory safety procedures and electrical safety requirements.
- Appropriate supervision and training are required.

Disclaimer of Liability

These design files, schematics, assembly guides, and associated materials are provided “AS IS”, without warranties of any kind, express or implied.

Users construct, modify, assemble, and operate this device at their own risk and are responsible for determining whether the device is appropriate and safe for their intended application.

To the extent permitted by applicable law, the designer and contributors assume no liability for injury, property damage, data loss, experimental loss, or other damages arising from fabrication, modification, assembly, operation, or use of the device.

EZ-Phoresis is intended for research and educational use and has not been validated or approved for clinical diagnostic use.

Overview

This document provides instructions to assemble the EZ-Phoresis system. Please ensure you have all supplies listed in the Bill of Materials before beginning assembly.

Important Information

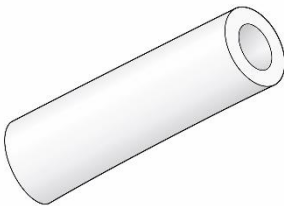
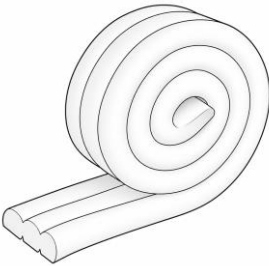
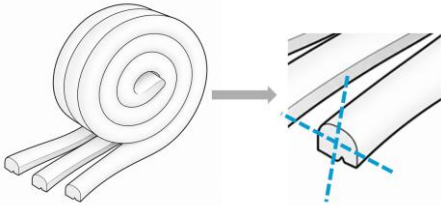
- This guide assumes that the user has a baseline understanding of 3D printing and gel electrophoresis
- The lid must be used at all times when current is running. Users **SHOULD NOT** attempt to touch the liquid buffer while current is running due to risk of electrical shock

Required Tools: Scissors, wire cutters, pliers (straight or needle nose), standard kitchen butter knife

3D Printing Guidance

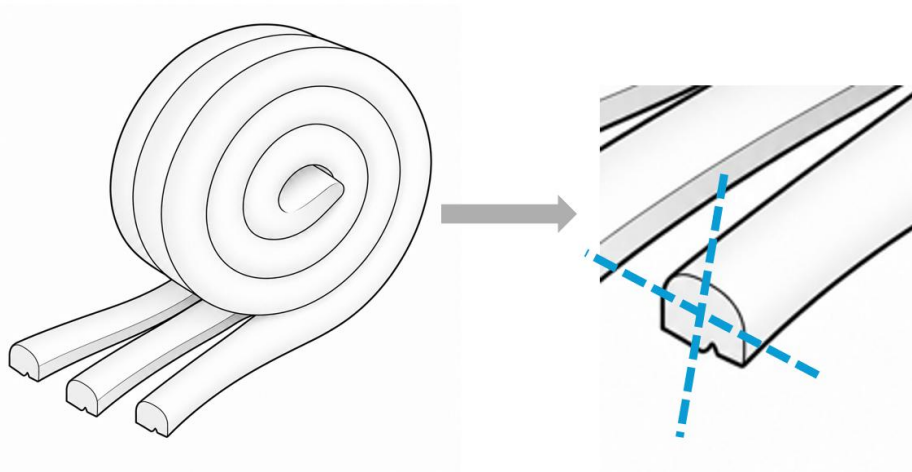
- Four Components will be printed via two separate STL files
 - **File 1:** Component A (Agarose gel casting tray) and Component B (Comb)
 - Note: The gel casting tray is designed to be used with painter's tape (or other similar material) to close the open sides. The gel is poured into the tray, and the tape is removed
 - **File 2:** Component C (Base) and Component D (Lid)
 - No assembly is required for components A, B, or D
- Filament selection
 - Any color can be chosen, but a light-colored or translucent filament may allow for easier visibility of bands during electrophoresis
 - This system was tested using PETG filament as Component C, and PLA for all other components
 - PLA may be sufficient for the base, though users may encounter less durability over time and worse permeability to water
- Use a 0.2mm layer height, 5(maximum)*** perimeters, 90% infill, and no structural supports

Materials for Assembly of EZ-Phoresis System (Component C)

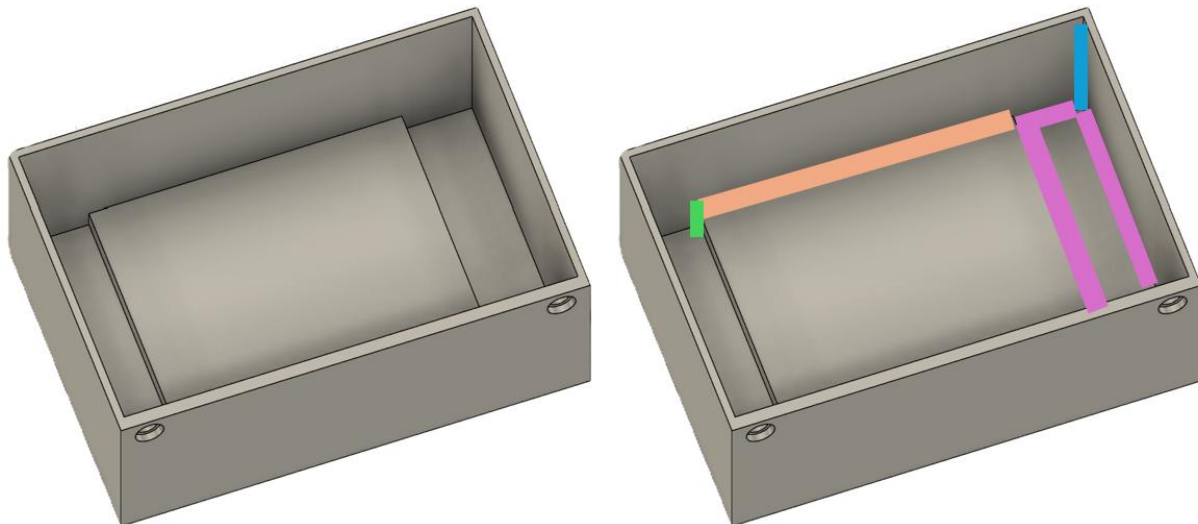
Image	Description	Number Required	Other Information
	PTFE Teflon tubing, 2mm ID x 4mm OD	x4 pieces, 4.5 cm each	
	22 gauge stainless steel wire	x2 pieces, 25 cm each	See Note *
	Female Banana Socket, solder-free	1 red, 1 black	 (Not Needed)
	Moldable silicone putty, white circular	 Cut strips into approximately ¼ cm length-wise strips	
<i>* Platinum wire is the standard material used in commercially available systems, and can also be used in the EZ-Phoresis system. However, stainless steel is more widely available and much more affordable. Users will need to replace stainless steel more often over time due to anodic dissolution during electrolysis</i>			

Assembly Instructions for EZ-Phoresis Base:

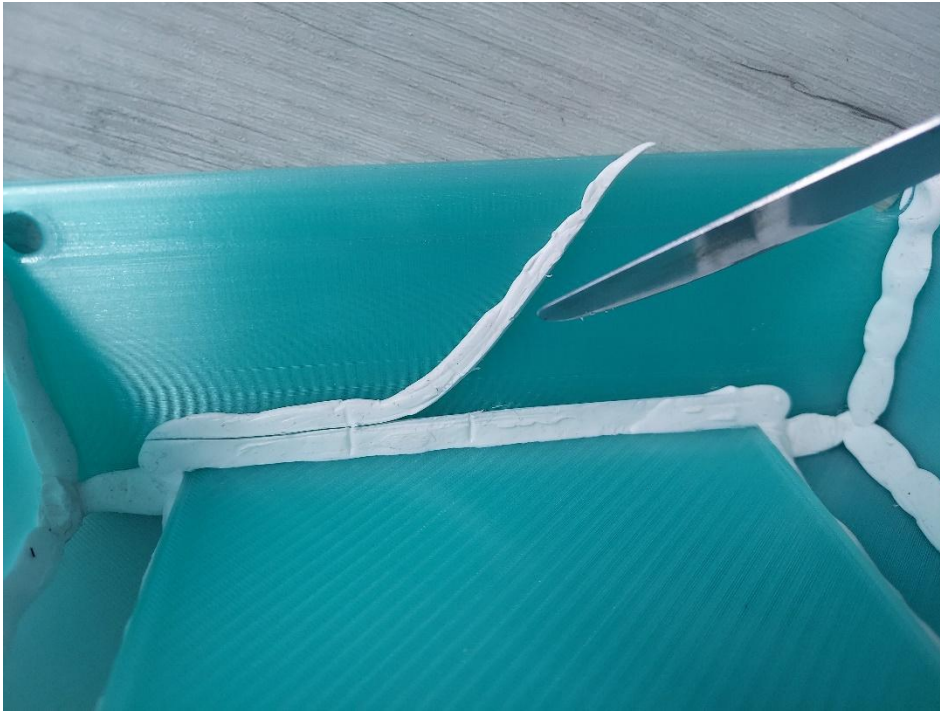
1. Create thin, long pieces of silicone putty by peeling the sections from the roll. Cut each section in quarters to create thin and long strips of silicon putty:



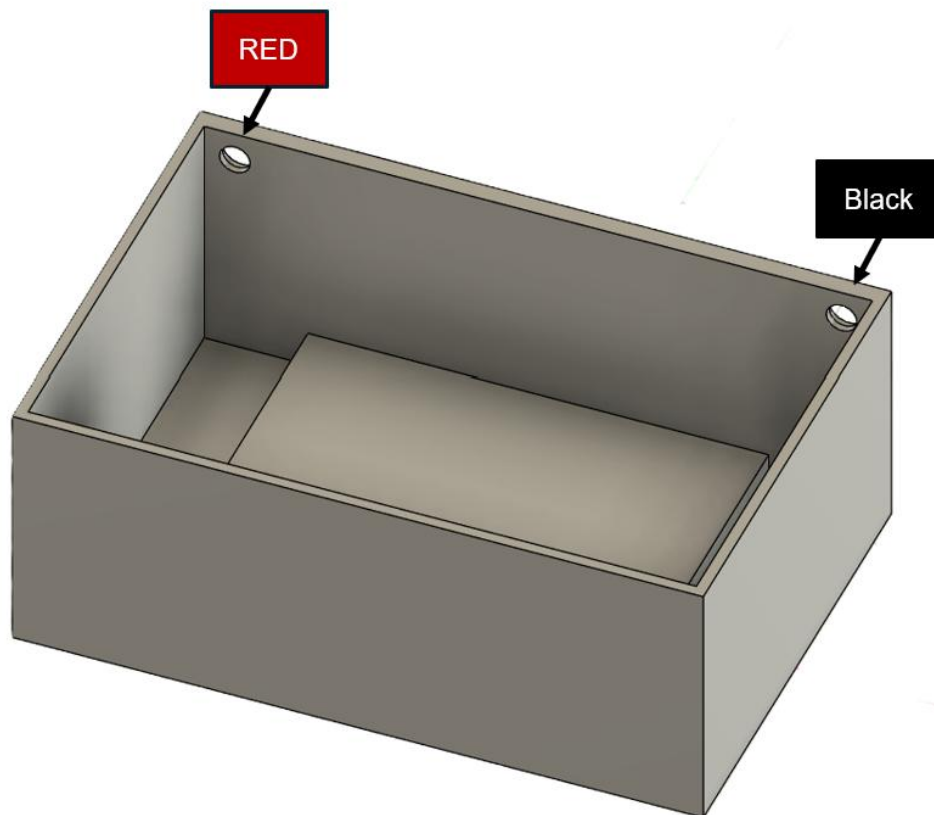
2. Use the silicon putty to insulate all inside corners. This will increase the impermeability of the base to liquids. We found that even after optimizing other factors, 3D printing did not consistently create a water-tight base. Silicon putty does not dry out and can be moved and re-applied as needed, offering a flexible solution
 - a. Four inside vertical corners (blue)
 - b. Two sections along the length of the box's sides where the gel tray will sit (orange) and the connecting vertical part to the box's base (green, 4 total))
 - c. Four sides of the box's base on each side (pink)



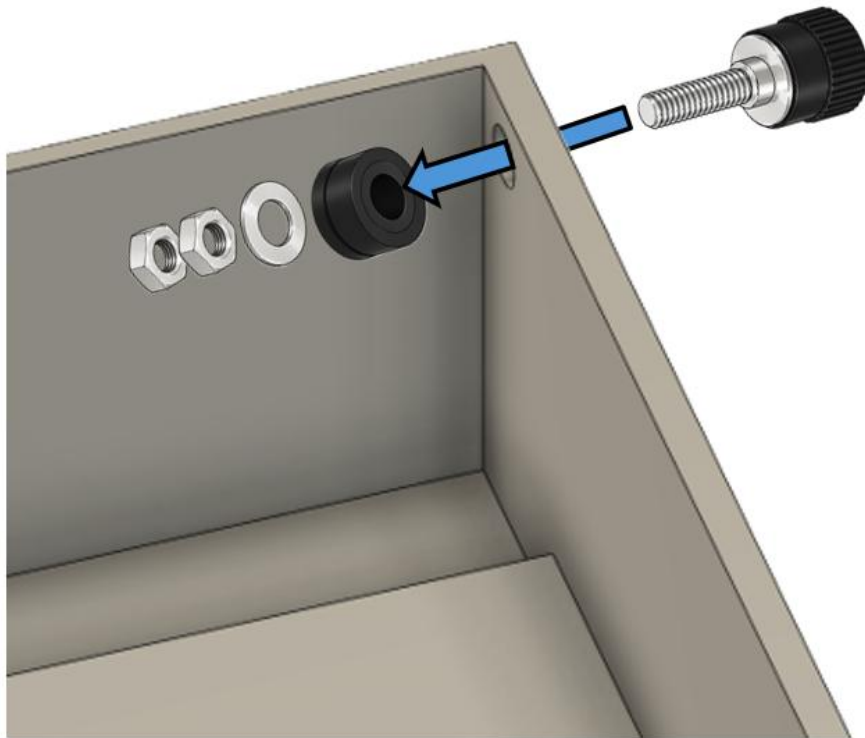
3. A flat kitchen knife can be used to create smooth edges (optional).



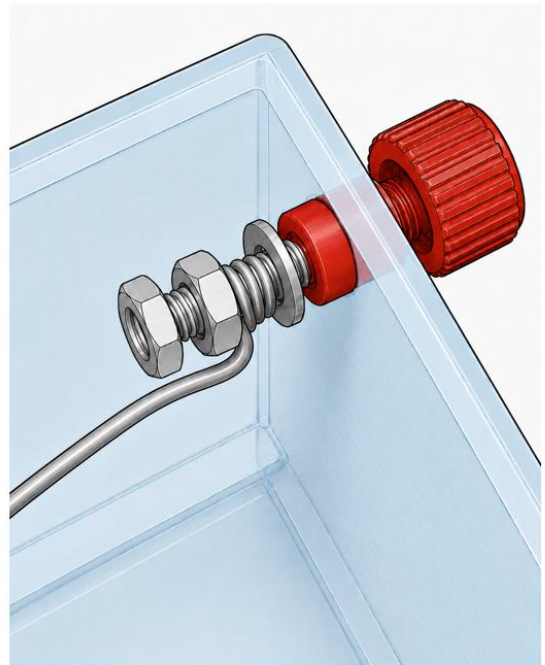
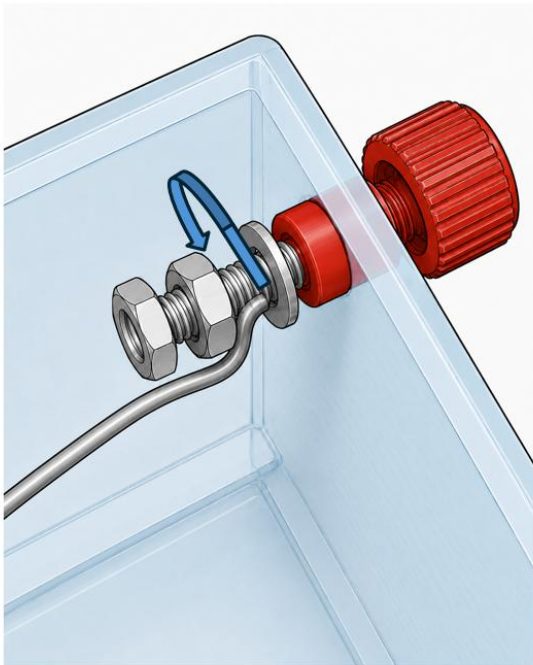
4. Attach the red and black banana plugs to the holes in Component C as follows:



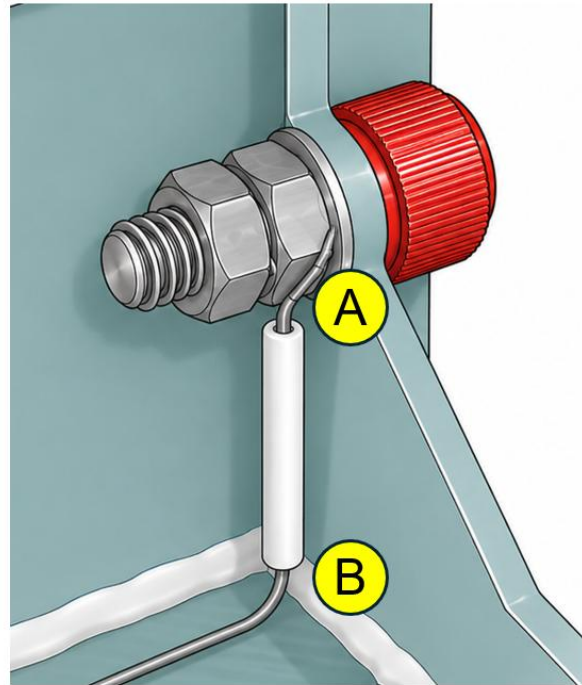
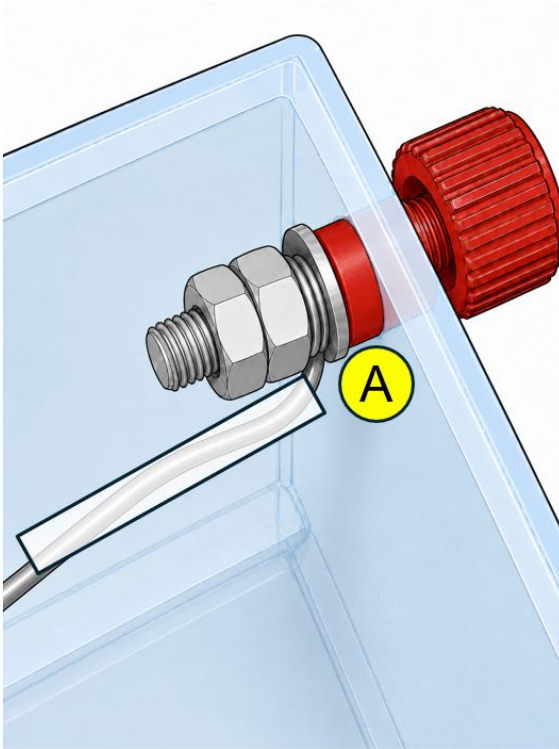
5. Loosely assemble the banana plug in the following order. The outermost piece will receive the electrical input device:



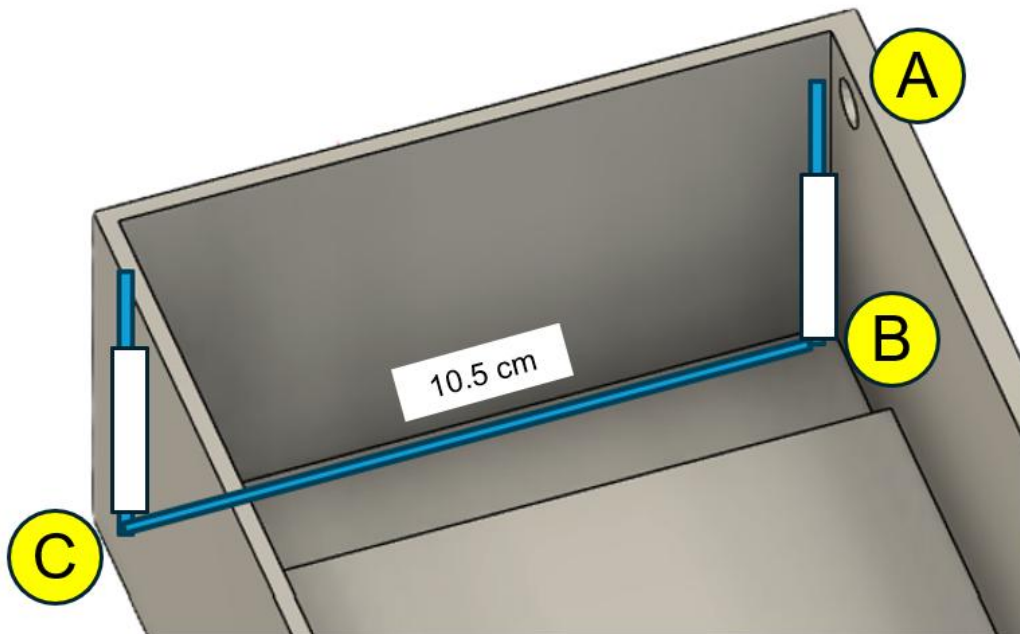
6. Use pliers to wrap the stainless steel wire 1-2 times between the washer and the first bolt:



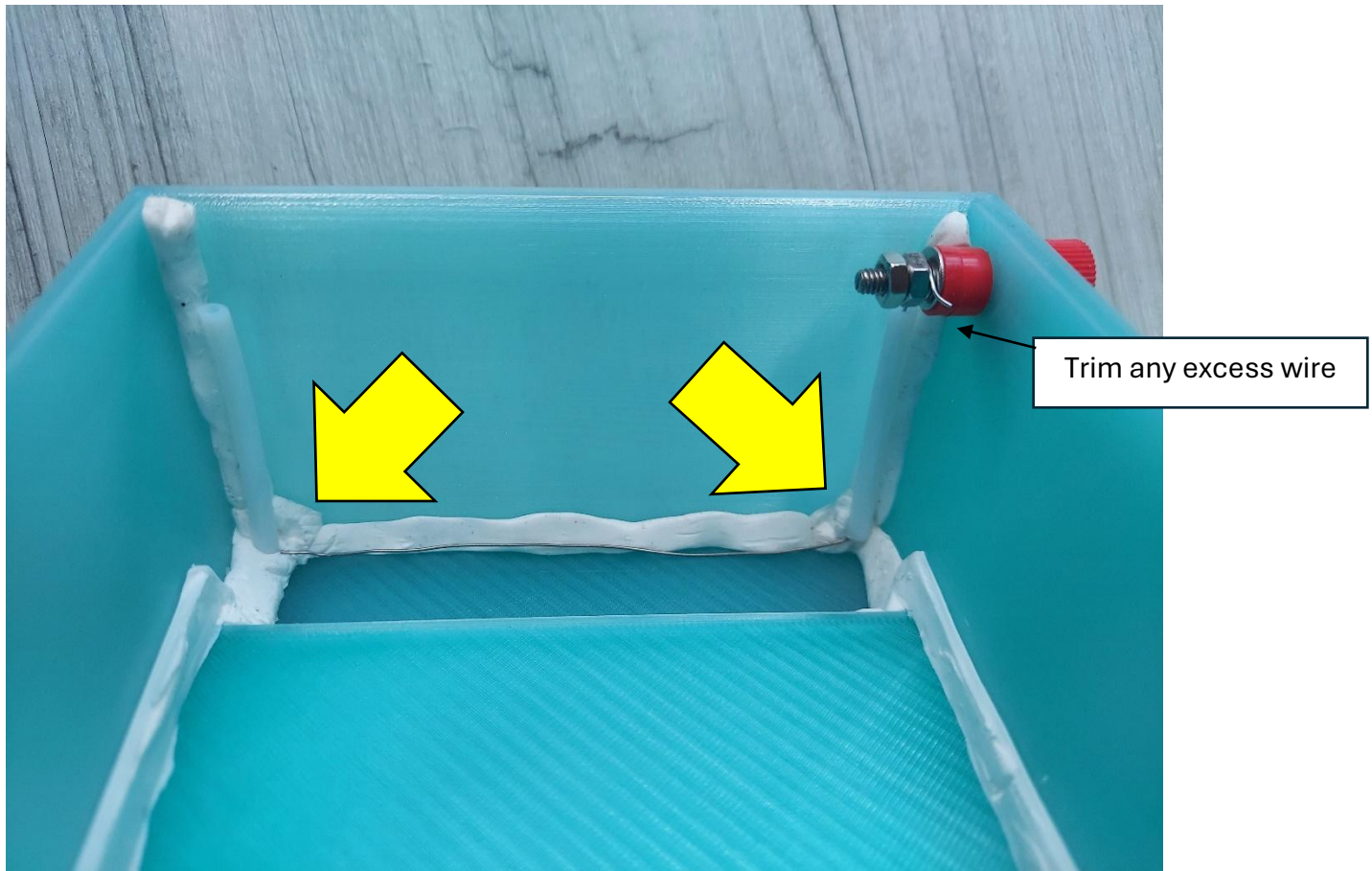
6. Tighten the nuts. Slide a piece of Teflon tubing onto the wire. Create a 90 degree bend in the wire (Point B) so that the wire runs parallel to the short edge of the box:



7. Measure 10.5cm and make another 90 degree bend at Point C. Slide another piece of Teflon over the vertical wire. You can trim any excess wire and adjust length as needed

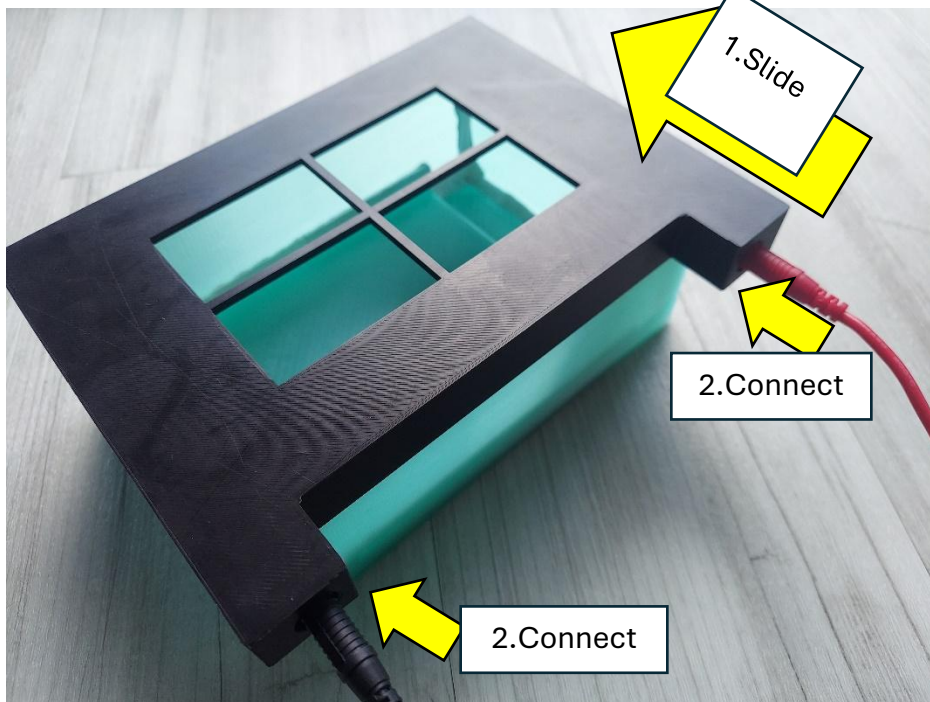


8. Use light pressure to secure the Teflon tubing into the silicone putty. The stainless steel wire can also be pushed into the putty for stability



9. Repeat on the other side, using the black banana plug

10. Attach the lid (Component D) by aligning the holes for the electrodes and sliding the lid on. Plug the electrodes through the holes and into the banana socket



Assembly is
Complete!